Bayesian SC vs Bayesian SDID

Geo-Lift Comparison on Simulated Panel

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June 18, 2026

1 What this project does

Inspired by a real-world migration in media-measurement platforms from Bayesian Synthetic Control (BSC) toward Bayesian Synthetic Difference-in-Differences (BSDID), this project fits both models on the same simulated panel with a known treatment effect and reports the difference. Results are shown for one representative simulated panel; a full Monte Carlo study would be needed to separate systematic bias from sampling noise.

2 Data

A 21-unit, 40-week panel. One treated unit, 20 controls. The treatment hits the treated unit at week 31 with a true per-week effect of $\tau = 25{,}000$ SEK. Each unit has its own baseline, a shared trend, annual seasonality, and Gaussian noise.

3 Model specification

3.1 Notation

Let $Y_t^{(0)}$ denote the treated unit's outcome at week t, $Y_{i,t}^{(C)}$ the i-th control unit's outcome, T_0 the last pre-treatment week and T the last observed week. Let $\omega \in \Delta^{N-1}$ (the N-simplex) be unit weights and $\lambda \in \Delta^{T_0-1}$ time weights.

3.2 Bayesian Synthetic Control (BSC)

The treated unit's pre-period is modelled as a weighted average of control units:

$$Y_t^{(0)} \sim \mathcal{N}\left(\sum_{i=1}^{N} \omega_i \, Y_{i,t}^{(C)},\ \sigma^2\right), \quad t = 1, \ldots, T_0$$

Priors:

$$\omega \sim \text{Dirichlet}(\mathbf{1}_N), \quad \sigma \sim \mathcal{N}^+(0,\, \widehat{\sigma}_{\text{emp}}^2)$$

where $\widehat{\sigma}_{\text{emp}} = \text{sd}(Y_{1:T_0}^{(0)})$ is the empirical pre-period standard deviation of the treated unit.

ATT (generated quantity): For each posterior draw s,

$$\text{counterfactual}_t^{(s)} = \sum_i \omega_i^{(s)} \, Y_{i,t}^{(C)}, \quad \text{ATT}_t^{(s)} = Y_t^{(0)} - \text{counterfactual}_t^{(s)}$$

for $t > T_0$. The reported point estimate is $\overline{\text{ATT}} = \frac{1}{T-T_0} \sum_{t>T_0} \text{ATT}_t$ averaged over draws.

Posterior:

$$p(\omega, \sigma \mid Y^{(0)}, Y^{(C)}) \ \propto\ \text{Dirichlet}(\omega \mid \mathbf{1}) \cdot \mathcal{N}^+(\sigma) \prod_{t=1}^{T_0} \mathcal{N}\left(Y_t^{(0)} \,\middle|\, \sum_i \omega_i Y_{i,t}^{(C)}, \sigma^2\right)$$

3.3 Bayesian Synthetic Difference-in-Differences (BSDID)

Same ω-weighted control layer as BSC, plus a second simplex λ of pre-period time weights. The ATT subtracts the λ-weighted pre-period gap:

$$\mathrm{ATT}_t = \left(Y_t^{(0)} - \sum_i \omega_i Y_{i,t}^{(C)}\right) - \sum_{s=1}^{T_0} \lambda_s \left(Y_s^{(0)} - \sum_i \omega_i Y_{i,s}^{(C)}\right)$$

Two likelihoods identify ω and λ jointly:

$$\text{Likelihood 1 (treated pre):} \quad Y_t^{(0)} \sim \mathcal{N}\left(\sum_i \omega_i Y_{i,t}^{(C)}, \sigma^2\right), \quad t = 1, \ldots, T_0$$

$$\text{Likelihood 2 (controls' pre-vs-post):} \quad \overline{Y_{i,\mathrm{post}}^{(C)}} \sim \mathcal{N}\left(\sum_t \lambda_t\, Y_{i,t}^{(C)},\ \sigma_\lambda^2\right), \quad i = 1, \ldots, N$$

where $\overline{Y_{i,\mathrm{post}}^{(C)}} = \frac{1}{T - T_0} \sum_{t > T_0} Y_{i,t}^{(C)}$.

Priors:

$$\omega \sim \mathrm{Dirichlet}(\mathbf{1}_N), \quad \lambda \sim \mathrm{Dirichlet}(\mathbf{1}_{T_0})$$
$$\sigma \sim \mathcal{N}^+(0,\, \widehat{\sigma}^2_{\mathrm{emp}}), \quad \sigma_\lambda \sim \mathcal{N}^+(0,\, \widehat{\sigma}^2_{\lambda,\mathrm{emp}})$$

where $\widehat{\sigma}_{\lambda,\mathrm{emp}}$ is the empirical standard deviation of $\overline{Y_{i,\mathrm{post}}^{(C)}}$ across controls.

Posterior:

$$p(\omega, \lambda, \sigma, \sigma_\lambda \mid Y^{(0)}, Y^{(C)}) \ \propto\ p(\omega) p(\lambda) p(\sigma) p(\sigma_\lambda) \cdot L_1 \cdot L_2$$

with L_1 and L_2 the two likelihoods above.

4 Inference

Both models fitted with Stan via cmdstanpy using HMC-NUTS. Diagnostics below.

Metric	**Value**
Chains	4
Warmup iterations	1 500 per chain
Sampling iterations	1 500 per chain
Post-warmup draws	6 000
max $\widehat{R}$	≈ 1.00
Divergences	0

Table 1: Convergence diagnostics for both BSC and BSDID fits.

5 Results

	Truth	BSC	BSDID
Mean ATT	25 000	19 550	29 333
90% interval	–	[14 386, 23 889]	[26 154, 32 301]
Interval width	–	9 503	**6 147**
|Error|	–	5 450	**4 333**

Table 2: ATT posterior summary on one simulated panel. BSDID has both a smaller absolute error and a 35% tighter interval than BSC. A Monte Carlo study over many panels would be required to separate systematic bias from sampling noise.

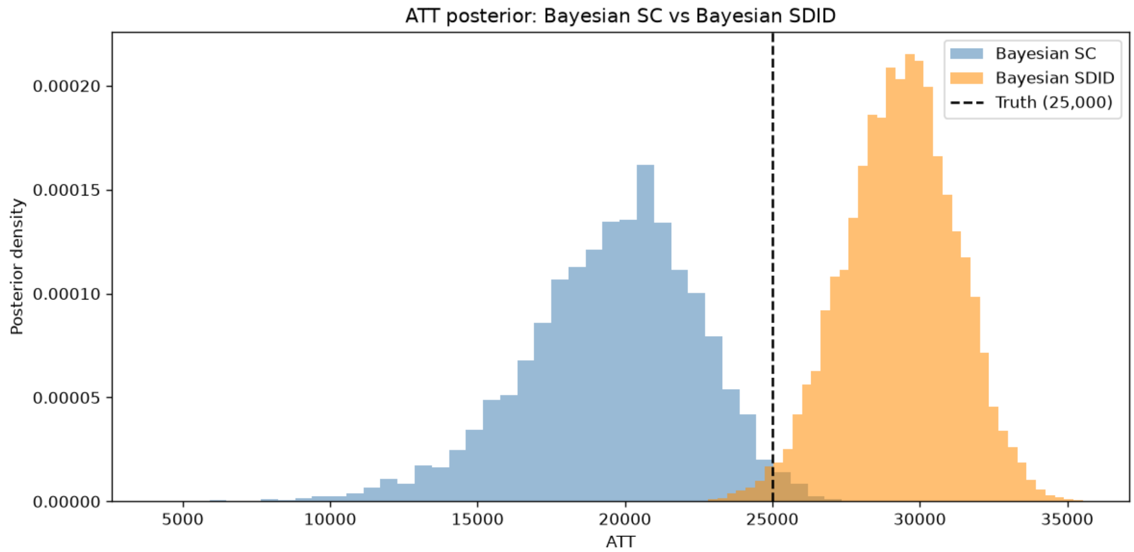

Figure 1: ATT posteriors. Dashed line is truth.

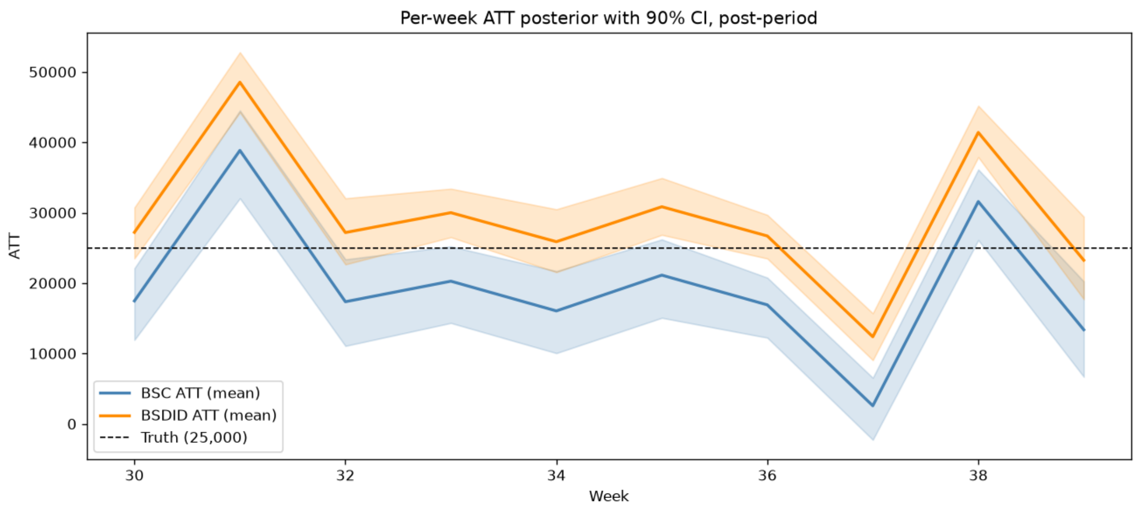

Figure 2: Per-week ATT, posterior mean and 90 % CI.

6 Why BSDID can outperform BSC

The treated unit sits slightly below its synthetic control in the pre-period. BSC reads that offset as part of the post-period treatment effect and underestimates τ by 22 %. BSDID subtracts the offset out of the ATT formula. On this panel it overshoots by 17 %, but with a tighter posterior and a smaller absolute error.

This is consistent with the motivation behind Arkhangelsky et al.'s SDID framework. It matters whenever the treated region happens to be a little above or below the convex hull of its controls in the pre-period, which is the common case on real geo data.

7 Limitations

This is a controlled method-comparison experiment where the ground truth must be known exactly, so several simplifications are deliberate: constant treatment effect, iid Gaussian noise, homogeneous control regions, and a generative structure close to what the models assume. The synthetic setup is cleaner than real geo data, which has non-stationarity, structural breaks, autocorrelated noise, regional shocks, and heteroskedasticity. Absolute numbers (35 % tighter interval, 20 % lower error) reflect this synthetic setup and would move on real data.

The results also come from a single random seed. A full Monte Carlo study over many seeds would be needed to separate systematic bias from sampling noise. Priors are uniform Dirichlet on both weight simplices, treatment is pure on/off, and there is no hierarchical pooling across experiments. Each of these is a natural extension for a follow-up study rather than a flaw in the current setup.

BSDID's theoretical advantage over BSC is that it subtracts a systematic pre-period offset between treated and synthetic control before computing the ATT. That advantage is most pronounced when the treated unit has a differential trend, structural drift, or persistent offset that BSC would misread as post-period treatment effect. The synthetic panel here has homogeneous trends across units and only a small pre-period offset on the treated unit, so BSDID's correction has relatively little to correct. On real geo data with differential regional trends and macro drift, BSDID's margin over BSC is expected to widen.